

İzmir Institute of Technology
MATH 142 Basic Calculus II - Midterm I

Name: _____

Student ID: _____

Department: _____

Instructor: _____ **SOLUTION KEY** _____

Exam list number: _____

Signature: _____

Duration: 105 Minutes

This exam contains 7 sheets (please check), including this sheet. Show your work in the space provided. You may use back of the pages. Please remark it at the end in case you use the back of a sheet.

- Your signature above simply means a lucid endorsement certifying that you haven't received any help during the exam. You ought to be aware of this and act accordingly.
- You are not allowed to use any book, note, smart phone, tablet or any calculator.
- A correct answer presented without explanation will receive no credit. Show all your work.
- You cannot leave the classroom before the first 30 minutes.
- You cannot use any statement out of scope of this exam.
- The exam consists of 6 questions of 20 points each, including 20 points bonus.
- Good luck.

Grade Table

Question:	1	2	3	4	5	6	Total
Points:	20	20	20	20	20	20	120
Score:							

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1. (20 points) The region bounded by the curves $y = x^3 - 2x$ and $y = x^2$ is revolved about the line $y = -3$ to generate a solid. Sketch the region and the solid, and find the volume of the solid.

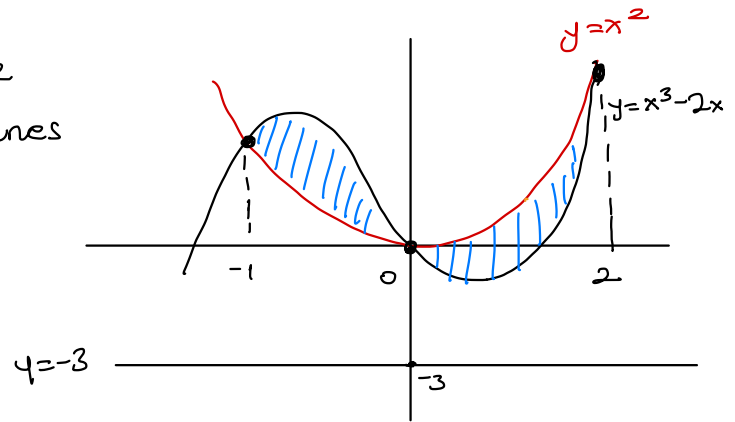
Solution : First, we need to find the intersection points of the curves $y = x^3 - 2x$ and $y = x^2$.

$$x^3 - 2x = x^2$$

$$\Rightarrow x^3 - x^2 - 2x = 0$$

$$\Rightarrow x(x+1)(x-2) = 0$$

$$\Rightarrow x = 0, -1, 2 \quad (\text{intersection points})$$



The volume is found by using the washer method around the line $y = -3$.

For $[-1, 0]$: Outer Radius: $R_1(x) = x^3 - 2x + 3$

Inner Radius : $r_1(x) = x^2 + 3$

For $[0, 2]$: Outer Radius : $R_2(x) : x^2 + 3$

Inner Radius : $r_2(x) : x^3 - 2x - (-3)$

$$V = \pi \int_{-1}^0 \left[(x^3 - 2x + 3)^2 - (x^2 + 3)^2 \right] dx + \pi \int_0^2 \left[(x^2 + 3)^2 - (x^3 - 2x - (-3))^2 \right] dx$$

$$= \pi \int_{-1}^0 \left[x^6 - 5x^4 + 6x^3 - 2x^2 + 12x \right] dx + \pi \int_0^2 \left[-x^6 + 5x^4 - 6x^3 + 2x^2 + 12x \right] dx$$

$$= \pi \left[\frac{x^7}{7} - x^5 + \frac{3}{2}x^4 - \frac{2}{3}x^3 - 6x^2 \right]_{-1}^0 + \pi \left[-\frac{x^7}{7} + x^5 - \frac{3}{2}x^4 + \frac{2}{3}x^3 + 6x^2 \right]_0^2$$

$$= \pi \left(0 - \left(-\frac{125}{42} \right) \right) + \pi \left(\frac{400}{21} - 0 \right)$$

$$= \frac{925\pi}{42}$$

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2. (20 points) Find the length of the curve $y = (x-1)^{2/3}$ from $x=0$ to $x=2$.

I. way

To find the arclength of the curve we will use arclength formula, $\int_a^b \sqrt{1+f'(x)^2} dx$.

We have $y = (x-1)^{2/3}$ from $x=0$ to $x=2$.

Let us express x as a function of y . $y = (x-1)^{2/3} \Rightarrow y^{3/2} = |x-1| \Rightarrow$ Function changes w.r.t. x .

So, we break the finding total arclength L into two parts. L_1 and L_2 .

For L_1 , we work on $0 \leq x \leq 1$ and our function is $y^{3/2} = -(x-1) \Rightarrow x_1 = 1 - y^{3/2}$, with $x=0 \Rightarrow y=1$ and $x=1 \Rightarrow y=0$.

$$\text{Then, } \frac{dx_1}{dy} = -\frac{3}{2} y^{1/2} \Rightarrow L_1 = \int_1^0 \sqrt{1 + \left(\frac{3}{2} y^{1/2}\right)^2} dy = - \int_0^1 \sqrt{1 + \frac{9}{4} y} dy \xrightarrow[u=\frac{9}{4}y]{u=1+\frac{9}{4}y} - \int_1^{13/4} \sqrt{u} \cdot \frac{4}{9} du = -\frac{4}{9} \cdot \left(\frac{2}{3} u^{3/2}\right) \Big|_1^{13/4}$$

$$\begin{matrix} y=0 \Rightarrow u=1 \\ y=1 \Rightarrow u=13/4 \end{matrix}$$

$$\Rightarrow L_1 = -\frac{8}{27} \cdot \left(\left(\frac{13}{4}\right)^{3/2} - 1\right) = -\frac{1}{27} \cdot (13\sqrt{13} - 8) \Rightarrow L_1 = \left| \frac{-1}{27} (13\sqrt{13} - 8) \right| = \frac{1}{27} (13\sqrt{13} - 8).$$

Similarly, for L_2 , we work on $1 \leq x \leq 2$ and our function is $x_2 = 1 + y^{3/2}$ with $x=1 \Rightarrow y=0$ and $x=2 \Rightarrow y=1$.

$$\text{Then, } \frac{dx_2}{dy} = \frac{3}{2} y^{1/2} \Rightarrow L_2 = \int_0^1 \sqrt{1 + \left(\frac{3}{2} y^{1/2}\right)^2} dy = \int_0^1 \sqrt{1 + \frac{9}{4} y} dy \xrightarrow[\text{we know from above}]{u=1+\frac{9}{4}y} \frac{(13\sqrt{13} - 8)}{27}.$$

$$\text{So, } L_2 = \frac{1}{27} \cdot (13\sqrt{13} - 8).$$

$$\text{Thus, our total arclength } L = L_1 + L_2 = \frac{2}{27} \cdot (13\sqrt{13} - 8).$$

Name: _____

2. (20 points) Find the length of the curve $y = (x-1)^{2/3}$ from $x = 0$ to $x = 2$.**II. way** To find the length of curve $y = (x-1)^{2/3}$, we will use arclength formula.

Arclength formula: $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \Rightarrow \frac{dy}{dx} = \frac{2}{3(x-1)^{1/3}} \Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{4}{9(x-1)^{2/3}}$

 $\Rightarrow$ Arclength from $x=0$ to $x=2$ is given by,

$$\text{Arc} = \int_0^2 \sqrt{1 + \frac{4}{9(x-1)^{2/3}}} dx = \int_0^2 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3|x-1|^{1/3}} dx = \int_0^2 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3|x-1|^{1/3}} dx.$$

As $|x-1| = 0$ at $x=1$, we separate the integral and use limit for improper integrals.

$$\text{Arc} = \int_0^1 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(1-x)^{1/3}} dx + \int_1^2 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(x-1)^{1/3}} dx = \lim_{a \rightarrow 1^-} \underbrace{\int_0^a \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(1-x)^{1/3}} dx}_{I_1} + \lim_{b \rightarrow 1^+} \underbrace{\int_b^2 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(x-1)^{1/3}} dx}_{I_2};$$

$$I_1 = \lim_{a \rightarrow 1^-} \int_0^a \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(1-x)^{1/3}} dx. \quad \text{Use substitution } u = 9(x-1)^{2/3} + 4, \quad du = \frac{2}{3}(x-1)^{-1/3} dx.$$

$$\Rightarrow du = \frac{2}{3}(x-1)^{-1/3} dx \Rightarrow \frac{-du}{6} = \frac{dx}{(1-x)^{1/3}} \quad \text{and } x=0 \Rightarrow u=13, \quad x \rightarrow 1^- \Rightarrow u \rightarrow 4^-.$$

$$\begin{aligned} \Rightarrow I_1 &= \lim_{a \rightarrow 4^-} \int_{13}^a \frac{\sqrt{u}}{3} \cdot \frac{du}{-6} = \lim_{a \rightarrow 4^-} \left(-\frac{1}{18} \cdot \left(\frac{2}{3} \cdot u^{3/2} \right) \Big|_{13}^a \right) = \lim_{a \rightarrow 4^-} -\frac{1}{18} \cdot \frac{2}{3} \left(a^{3/2} - 13^{3/2} \right) = -\frac{1}{27} \cdot \left(4^{3/2} - 13^{3/2} \right) \\ &= \frac{13\sqrt{13} - 8}{27} // \end{aligned}$$

Similarly, $I_2 = \lim_{b \rightarrow 1^+} \int_b^2 \frac{\sqrt{9(x-1)^{2/3} + 4}}{3(x-1)^{1/3}} dx.$ Substitute $u = 9(x-1)^{2/3} + 4 \Rightarrow$ we get that
 $\begin{pmatrix} x=2, u=13 \\ x \rightarrow 1^+, u \rightarrow 4^+ \end{pmatrix}$

$$I_2 = \lim_{b \rightarrow 4^+} \int_{13}^b \frac{\sqrt{u}}{3} \cdot \frac{du}{6} = \lim_{b \rightarrow 4^+} \frac{1}{18} \cdot \frac{2}{3} \cdot \left(u^{3/2} \right) \Big|_{13}^b = \frac{1}{27} \cdot \left(13^{3/2} - 4^{3/2} \right) = \frac{13\sqrt{13} - 8}{27}.$$

$$\text{Thus, Arc} = I_1 + I_2 = \frac{2}{27} \cdot (13\sqrt{13} - 8).$$

Name: Solution Key

3. Evaluate the following integrals:

(a) (10 points) $\int z(\ln z)^2 dz$ (*)

(b) (10 points) $\int_1^{\sqrt{3}} \frac{3t^2 + t + 4}{t^3 + t} dt$

Solution.**a.** We illustrate two distinct approaches to evaluate the integral (*):**1st way:**

We use integration by parts. Let

$$u = (\ln z)^2 \quad dv = z dz$$

$$du = 2 \ln z \cdot \frac{1}{z} dz \quad v = \frac{z^2}{2}$$

and so

$$\int z(\ln z)^2 dz = \frac{z^2}{2}(\ln z)^2 - \int \frac{z^2}{2} \cdot 2 \ln z \cdot \frac{1}{z} dz$$

$$= \frac{z^2}{2}(\ln z)^2 - \underbrace{\int z \ln z dz}_I$$

Say $I = \int z \ln z dz$. To compute I , we use again integration by parts:

$$\begin{cases} u = \ln z & dv = z dz, \\ du = \frac{1}{z} dz & v = \frac{z^2}{2} \end{cases}$$

and then

$$I = \frac{z^2}{2}(\ln z) - \int \frac{z^2}{2} \cdot \frac{1}{z} dz,$$

$$= \frac{z^2}{2} \ln z - \int \frac{z}{2} dz,$$

$$= \frac{z^2}{2} \ln z - \frac{z^2}{4} + C,$$

for some constant C . Substituting back yields

$$\int z(\ln z)^2 dz = \frac{z^2}{2}(\ln z)^2 - \left(\frac{z^2}{2} \ln z - \frac{z^2}{4} + C \right)$$

$$= \frac{z^2}{2} \left((\ln z)^2 - \ln z \right) + \frac{z^2}{4} - C$$

$$= \frac{z^2}{2} \left((\ln z)^2 - \ln z + \frac{1}{2} \right) + C_1,$$

for some constant C_1 .

2nd way: By using the following substitution

$$\begin{cases} t = \ln x & \Rightarrow x = e^t \\ dt = \frac{1}{x} dx & \Rightarrow x dt = dx \Rightarrow e^t dt = dx, \end{cases}$$

the integral (*) becomes

$$\int e^t \cdot t^2 \cdot e^t dt = \int t^2 \cdot e^{2t} dt.$$

Now we use integration by parts. Let

$$\begin{cases} u = t^2 & dv = e^{2t} dt, \\ du = 2t dt & v = \frac{e^{2t}}{2} \end{cases}$$

and then we get

$$\int t^2 e^{2t} dt = \frac{t^2 \cdot e^{2t}}{2} - \int 2t \cdot \frac{e^{2t}}{2} dt,$$

$$\int t^2 e^{2t} dt = \frac{t^2 \cdot e^{2t}}{2} - \int t \cdot e^{2t} dt.$$

Say $I = \int t e^{2t} dt$. To compute the integral I , we use again integration by parts. Let

$$\begin{cases} u = t & dv = e^{2t} dt \\ du = dt & v = \frac{e^{2t}}{2} \end{cases}$$

and so we have

$$\begin{aligned} I &= \frac{t \cdot e^{2t}}{2} - \int \frac{e^{2t}}{2} dt \\ &= \frac{t \cdot e^{2t}}{2} - \frac{e^{2t}}{4} + c \end{aligned}$$

for some constant c . Combining the results yields

$$\int t^2 e^{2t} dt = \frac{t^2 \cdot e^{2t}}{2} - \int t \cdot e^{2t} dt$$

$$= \frac{t^2}{2} e^{2t} - \left(\frac{t}{2} e^{2t} - \frac{e^{2t}}{4} + c \right)$$

$$= \frac{t^2}{2} e^{2t} - \frac{t}{2} e^{2t} + \frac{e^{2t}}{4} - c$$

Then we get

$$\int t^2 e^{2t} dt = \frac{e^{2t}}{2} \left(t^2 - t + \frac{1}{2} \right) + C_1$$

for some constant C_1 . With back substitution ($t = \ln x$ and so $e^t = x$, i.e., $(e^t)^2 = e^{2t} = x^2$), the integral (*) becomes

$$\int x (\ln x)^2 dx = \frac{x^2}{2} \left((\ln x)^2 - \ln x + \frac{1}{2} \right) + C_1.$$

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b. By partial fraction, we can write the integrand as

$$\frac{3t^2+t+4}{t(t^2+1)} = \frac{A}{t} + \frac{Bt+C}{t^2+1},$$

then we try to find the coefficients A, B and C by solving the following

$$A(t^2+1) + (Bt+C)t = 3t^2+t+4,$$

$$t^2(A+B) + t(C) + 1 \cdot (A) = 3t^2+t+4.$$

So the coefficients become

$$A+B=3, \quad C=1, \quad A=4 \quad \text{and so} \quad B=-1.$$

Therefore, we have

$$\int_1^{\sqrt{3}} \frac{3t^2+t+4}{t^3+t} dt = \int_1^{\sqrt{3}} \left[\frac{4}{t} + \frac{-t+1}{t^2+1} \right] dt$$

$$= \int_1^{\sqrt{3}} \frac{4}{t} dt - \int_1^{\sqrt{3}} \frac{t}{t^2+1} dt + \int_1^{\sqrt{3}} \frac{1}{t^2+1} dt, \quad \text{by the substitution}$$

$$\begin{cases} u=t^2+1 \\ du=2t dt \end{cases} \Rightarrow \begin{cases} t=1 \Rightarrow u=2 \\ t=\sqrt{3} \Rightarrow u=4 \end{cases}$$

$$= 4 \ln|t| \Big|_1^{\sqrt{3}} - \frac{1}{2} \int_2^4 \frac{du}{u} + \arctan t \Big|_1^{\sqrt{3}}$$

$$= 4 (\ln \sqrt{3} - \ln 1) - \frac{1}{2} (\ln|u| \Big|_2^4) + (\arctan \sqrt{3} - \arctan 1),$$

$$= 4 \ln(3^{1/2}) - \frac{1}{2} (\ln 4 - \ln 2) + \left(\frac{\pi}{3} - \frac{\pi}{4} \right),$$

$$= 4 \cdot \frac{1}{2} \ln(3) - \frac{1}{2} \left(\ln \frac{4}{2} \right) + \frac{\pi}{12} \quad \text{by the property of logarithms.}$$

Then we get

$$\int_1^{\sqrt{3}} \frac{3t^2+t+4}{t^3+t} dt = 2 \ln 3 - \frac{1}{2} \ln 2 + \frac{\pi}{12}.$$

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4. Evaluate the following integrals:

(a) (10 points) $\int \frac{\sqrt{x}}{\sqrt{1-x^3}} dx$

(b) (10 points) $\int \frac{d\theta}{1-\cos\theta}$

a) Let us substitute $u = x^{3/2}$, then we have $du = \frac{3}{2} \cdot x^{1/2} dx$. So, our integral becomes

$$\int \frac{\sqrt{x}}{\sqrt{1-x^3}} dx = \int \frac{1}{\sqrt{1-u^2}} \cdot \frac{2}{3} du = \frac{2}{3} \int \frac{du}{\sqrt{1-u^2}}.$$

Again using another substitution $u = \sin\theta$, $du = \cos\theta d\theta$, with $-\pi/2 \leq \theta \leq \pi/2$, we obtain the following

$$\int \frac{du}{\sqrt{1-u^2}} = \int \frac{\cos\theta d\theta}{\sqrt{1-\sin^2\theta}} = \int \frac{\cos\theta d\theta}{\sqrt{\cos^2\theta}} = \int \frac{\cos\theta}{|\cos\theta|} d\theta \quad \text{since } \theta \in [-\pi/2, \pi/2] \quad \int \frac{\cos\theta}{\cos\theta} d\theta = \int 1 d\theta = \theta + C.$$

Substitute back, we get $\int \frac{du}{\sqrt{1-u^2}} = \arcsin(u) + C.$ Thus, $\int \frac{\sqrt{x}}{\sqrt{1-x^3}} dx = \frac{2}{3} \arcsin(x^{3/2}) + C.$ b) We will use the trigonometric half identity $1-\cos\theta = 2\sin^2(\frac{\theta}{2})$.Then, our integral becomes $\int \frac{d\theta}{1-\cos\theta} = \int \frac{d\theta}{2\sin^2(\frac{\theta}{2})} = \frac{1}{2} \int \csc^2(\frac{\theta}{2}) d\theta.$ Now, let us use substitution $u = \frac{\theta}{2}$, $du = \frac{d\theta}{2}$. Then we get $\frac{1}{2} \int \csc^2(\frac{\theta}{2}) d\theta = \int \csc^2(u) du.$ Lastly, we know that $\int \csc^2 u du = -\cot(u) + C$. Thus, $\int \frac{d\theta}{1-\cos\theta} = -\cot(\frac{\theta}{2}) + C.$ Another way
Multiply and divide $\frac{1}{1-\cos\theta}$ by $(1+\cos\theta)$. Then we get $\frac{1+\cos\theta}{1-\cos^2\theta}.$

$$\text{So, } \int \frac{d\theta}{1-\cos\theta} = \int \frac{1+\cos\theta}{1-\cos^2\theta} d\theta = \int \frac{1+\cos\theta}{\sin^2\theta} d\theta = \int \frac{d\theta}{\sin^2\theta} + \int \frac{\cos\theta}{\sin^2\theta} d\theta = \int \csc^2\theta d\theta + \int \frac{du}{u^2}$$

Substitute $u = \sin\theta$,
 $du = \cos\theta d\theta$

$$\text{Thus, } \int \frac{d\theta}{1-\cos\theta} = -\cot(\theta) - \frac{1}{u} + C = -\cot(\theta) - \frac{1}{\sin\theta} + C = -\cot(\theta/2) + C.$$

$$\left[-\cot\theta - \frac{1}{\sin\theta} = \frac{-(1+\cos\theta)}{\sin\theta} = \frac{-(2\cos^2(\theta/2))}{2\sin(\theta/2)\cos(\theta/2)} = -\cot(\theta/2). \right]$$

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5. (a) (10 points) Evaluate the following integral

$$\int_0^2 \frac{dx}{\sqrt{|x-1|}}$$

- (b) (10 points) Determine whether the convergency of the following integral

$$\int_2^\infty \frac{\sqrt{x^2+1}}{x^2+\ln(x)} dx$$

a) As $|x-1| = \begin{cases} x-1, & \text{if } x \geq 1 \\ 1-x, & \text{if } x < 1 \end{cases}$,

we need to split $\int_0^2 \frac{dx}{\sqrt{|x-1|}}$ as $\int_0^1 \frac{dx}{\sqrt{1-x}} + \int_1^2 \frac{dx}{\sqrt{x-1}}$

Since $\frac{1}{\sqrt{1-x}}$ and $\frac{1}{\sqrt{x-1}}$ are undefined at $x=1$, the

integrals

$$\int_0^1 \frac{dx}{\sqrt{1-x}} + \int_1^2 \frac{dx}{\sqrt{x-1}}$$

are improper. Thus,

$$\int_0^1 \frac{dx}{\sqrt{1-x}} + \int_1^2 \frac{dx}{\sqrt{x-1}} = \lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{\sqrt{1-x}} + \lim_{c \rightarrow 1^+} \int_c^2 \frac{dx}{\sqrt{x-1}},$$

where

$$\int_0^b \frac{dx}{\sqrt{1-x}} = - \int \frac{du}{\sqrt{u}} = -2\sqrt{u} = -2\sqrt{1-x} \Big|_0^b = -2\sqrt{1-b} + 2,$$

$$\begin{aligned} u &= 1-x \\ \Rightarrow du &= -dx \end{aligned}$$

$$\int_c^2 \frac{dx}{\sqrt{x-1}} = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} = 2\sqrt{x-1} \Big|_c^2 = 2 - 2\sqrt{c-1}.$$

$$\begin{aligned} t &= x-1 \\ \Rightarrow dt &= dx \end{aligned}$$

Thus,

$$\int_0^2 \frac{dx}{\sqrt{|x-1|}} = \lim_{b \rightarrow 1^-} (-2\sqrt{1-b} + 2) + \lim_{c \rightarrow 1^+} (2 - 2\sqrt{c-1}) = 2 + 2 = 4$$

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5. (a) (10 points) Evaluate the following integral

$$\int_0^2 \frac{dx}{\sqrt{|x-1|}}$$

- (b) (10 points) Determine whether the convergency of the following integral

$$\int_2^{\infty} \frac{\sqrt{x^2+1}}{x^2+\ln(x)} dx$$

$$b) \int_2^{\infty} \frac{\sqrt{x^2+1}}{x^2+\ln x} dx$$

For large x , $\sqrt{x^2+1} \sim x$, $x^2+\ln(x) \sim x^2$.

$$\Rightarrow \frac{\sqrt{x^2+1}}{x^2+\ln x} \sim \frac{x}{x^2} = \frac{1}{x}.$$

Thus, let us compare the integral with $\int_2^{\infty} \frac{1}{x} dx$, which

diverges.

$$\text{Let } f(x) := \frac{\sqrt{x^2+1}}{x^2+\ln x}, \quad g(x) := \frac{1}{x}.$$

$$\begin{aligned} \text{Then } \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow \infty} \frac{x\sqrt{x^2+1}}{x^2+\ln x} = \lim_{x \rightarrow \infty} \frac{x\sqrt{x^2(1+\frac{1}{x^2})}}{x^2+\ln x} \\ &= \lim_{x \rightarrow \infty} \frac{x^2\sqrt{1+\frac{1}{x^2}}}{x^2(1+\frac{\ln x}{x^2})} = 1 > 0. \end{aligned}$$

By the limit comparison test, since $\int_2^{\infty} \frac{1}{x} dx$ diverges,

$$\int_2^{\infty} \frac{\sqrt{x^2+1}}{x^2+\ln x} dx \text{ diverges, as well.}$$

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6. Which of the following series converge, and which diverge? Give reasons for your answers. If a series converges, find its sum.

(a) (10 points) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n}$

(b) (10 points) $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

$$a) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n} = \sum_{n=1}^{\infty} (-1)^n \cdot -1 \cdot \frac{3}{2^n} = -3 \sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n,$$

where

$$\sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n \text{ is a geometric series.}$$

$$\text{As } \left| -\frac{1}{2} \right| = \frac{1}{2} < 1, \sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n \text{ converges, and}$$

$$\sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n = \frac{-1/2}{1 - (-1/2)} = \frac{-1/2}{3/2} = -\frac{1}{3}$$

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{3}{2^n} = -3 \cdot -\frac{1}{3} = 1$$

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6. Which of the following series converge, and which diverge? Give reasons for your answers. If a series converges, find its sum.

(a) (10 points) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n}$

(b) (10 points) $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

b) $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

Let $S_N := \sum_{n=0}^N \cos\left(\frac{n\pi}{2}\right)$.

Observe that

$$S_0 = 1$$

$$S_1 = 1 + 0 = 1$$

$$S_2 = 1 + 0 - 1 = 0$$

$$S_3 = 1 + 0 - 1 + 0 = 0$$

$$S_4 = 1 + 0 - 1 + 0 + 1 = 1$$

$$S_5 = 1 + 0 - 1 + 0 + 1 + 0 = 1$$

$$S_6 = 1 + 0 - 1 + 0 + 1 + 0 - 1 = 0$$

$$S_7 = 1 + 0 - 1 + 0 + 1 + 0 - 1 + 0 = 0$$

In general,
$$S_N = \begin{cases} 1, & \text{if } N = 4k \text{ or } 4k+1 \\ 0, & \text{if } N = 4k+2 \text{ or } 4k+3 \end{cases}$$

$\Rightarrow \lim_{N \rightarrow \infty} S_N$ does not exist.

So, $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$ diverges.